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GaussianDWM: 3D Gaussian Driving World Model for Unified Scene Understanding and Multi-Modal Generation

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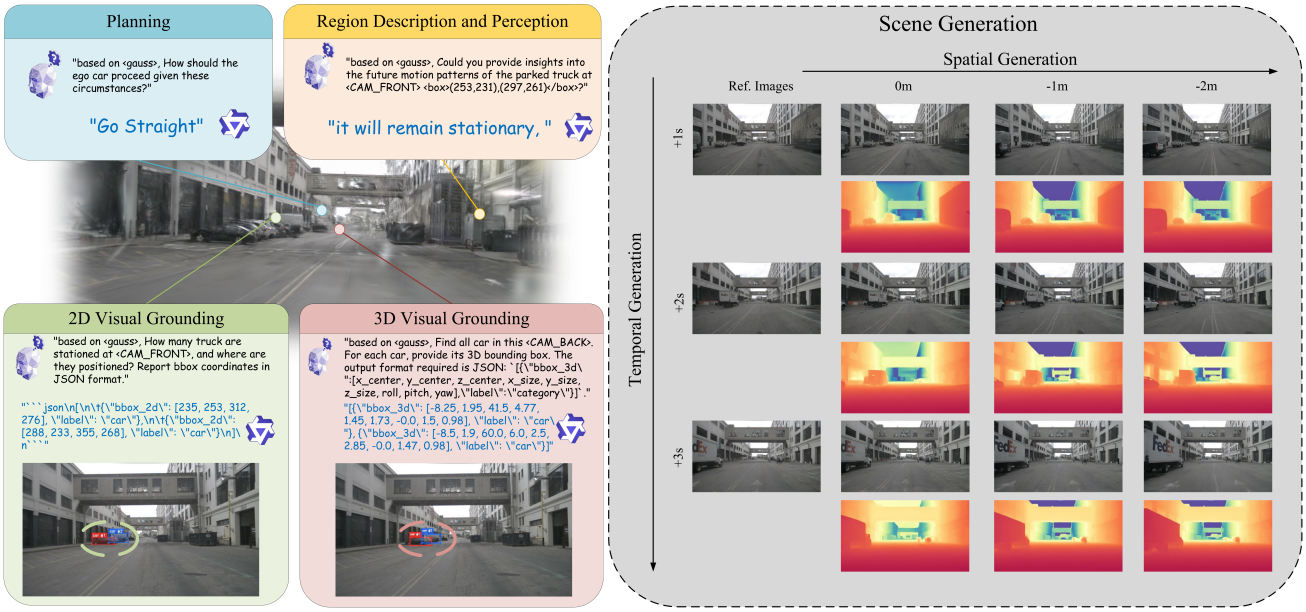


Figure 1. We propose the first unified 3D Gaussian-based world model framework that achieves comprehensive scene understanding and scene generation for driving scenarios. It efficiently encodes complex scenes, samples task-relevant information, and handles diverse question-answering tasks. Moreover, by leveraging the extracted world knowledge, our framework guides the generative model to perform accurate spatial and temporal scene generation.

Abstract

Driving World Models (DWMs) have been developing rapidly with the advances of generative models. However, existing DWMs lack 3D scene understanding capabilities and can only generate content conditioned on input data, without the ability to interpret or reason about the driving environment. Moreover, current approaches represent 3D spatial information with point cloud or BEV features do not accurately align textual information with the underlying 3D scene. To address these limitations, we propose a novel unified DWM framework based on 3D Gaussian scene representation, which enables both 3D scene understanding and multi-modal scene generation, while also enabling contextual enrichment for understanding and generation tasks. Our approach directly aligns textual information with the 3D scene by embedding rich linguistic features into each Gaussian primitive, thereby achieving early modality alignment. In addition, we design a novel task-aware language-guided sampling strategy that removes redundant 3D Gaussians and injects accurate and compact 3D tokens into LLM. Furthermore, we design a dual-condition multi-modal generation model, where the information captured by our vision-language model is leveraged as a high-level language condition in combination with a low-level image condition, jointly guiding the multi-modal generation process. We conduct comprehensive studies on the nuScenes, and NuInteract datasets to validate the effectiveness of our framework. Our method achieves state-of-the-art performance. We will release the code publicly on

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GitHub https://github.com/dtc111111/GaussianDWM .

1. Introduction

Driving World Model (DWM) [3, 20, 28, 57, 63] have become essential for autonomous driving for their ability to predict future scene generation. These models predict environmental changes and synthesize simulation data for risk forecasting, route optimization, and corner case training. Most existing approaches achieve this by predicting modalities such as images [70, 76] and point clouds [36, 37, 74], which represent the visual and geometric properties of the environment. While these DWMs excel at forecasting how the environment may evolve, they are difficult to interpret, describe, or query, and cannot easily provide contextual information (e.g., visual question answering or scene description). With the rapid progress of Large-Language Models and Vision-Language Models (VLMs) [56, 59–61, 69, 81], remarkable advancements have been achieved in general vision tasks by leveraging world knowledge and causal reasoning. This highlights the potential of combining the scene understanding capability of VLMs with the generative power of DWMs as a promising future direction. Notably, pioneering efforts such as HERMES [80] and UniFuture [38] have first achieved the unification of scene generation and understanding for autonomous driving world models. They adopts BEV/Depth features to represent spatial information, aligns them with the text space, and incorporates them into the generative model. However, this BEV-based scene representation only achieves feature-level alignment between textual and spatial inputs, which is not sufficiently accurate. To overcome this limitation, we propose a novel 3DGS-based scene representation that unifies scene understanding and generation.

First, we directly embed linguistic features into the 3D Gaussians, thereby achieving explicit spatial alignment between text and the 3D scene. This improves the accuracy of cross-modal alignment. Second, given the redundancy of Gaussians (i.e., extremely dense representations with tens of thousands of tokens per scene), it is impractical for VLMs to process such a large number of tokens effectively. To address this, we introduce a novel task-aware language-guided sampling strategy. By calculating the similarity between the input text and the 3D Gaussians, our method selects the most relevant Gaussians for the query and projects them into the context space through our projector, injecting accurate and compact 3D tokens into the textual understanding. Finally, we design a dual-condition multi-modal generation model. In this framework, the understanding from the VLM provides a high-level language condition, while image features serve as a low-level image condition. Together, they guide the generation of multiple modalities, including RGB, depth, and language. Moreover, our framework supports both spatial and temporal generation. Overall, our contributions are shown as follows:

- We propose the first 3D Gaussian-based unified world model framework that supports both scene understanding and scene generation.
- We introduce a novel token extraction and projection module for 3D Gaussian scene representations. Due to the redundancy of 3D Gaussians, we further develop a task-aware language-guided sampling strategy that overcomes token length limitations while preserving essential spatial information.
- We design a novel dual-condition multi-modal scene generation framework with high-level feature from world knowledge and low-level feature from images.
- Extended experiments on the Nuscenes and NuInteract datasets demonstrate that our method effectively bridges the gap between understanding and generation, enabling both accurate scene comprehension and more coherent future scene prediction.

2. Related Work

Novel View Synthesis for Urban Scene With the emergence of NeRF [44] and 3D Gaussian Splatting (3DGS) [30], many methods have adopted these representations across a wide range of tasks, including robotic mapping and localization [9, 11, 12, 14–16], VR [13, 21–23, 34, 83], and autonomous driving [5, 45]. Early NeRF-based methods such as NSG [45], SUDS [53], ProSGNeRF [8], EmerNeRF [71], and FreeDriveRF [62] achieved dynamic–static disentangled reconstruction through the use of scene graphs, optical flow, or other motion cues. More recently, several 3DGS-based approaches have been proposed to further improve rendering efficiency. PVG [4] introduces periodic vibration-based temporal dynamics to unify static and dynamic elements without manual annotations. Methods such as Street Gaussians [68], Driving Gaussian [79], and DeSiRe-GS [48] also explicitly separate dynamic and static components for reconstruction. LESSON [47] proposes a teacher-guided diffusion strategy for generating 3D Gaussian splats using only 2D supervision. STORM [72] proposes a feed-forward Transformer architecture to infer dynamic Gaussians and their velocities, enabling efficient large-scale outdoor scene reconstruction. DrivingForward [51] achieves feed-forward reconstruction from sparse surround-view inputs using self-supervised pose and depth estimation. MUDG [84] and Dist-4D [25] propose multi-modal novel view synthesis frameworks that generate both RGB and depth modalities.

Driving World Model Driving world models [18, 32, 40, 52, 67] have attracted considerable attention in autonomous driving due to their ability to provide comprehensive environmental representations and predict future scenarios. Current methods primarily rely on 2D and 3D conditions

for scene generation. GAIA-1 [29] introduces an autoregressive model for video generation in driving scenarios. DriveDreamer [76] proposes a scene generation framework conditioned on 3D structure to provide geometric representations that benefit downstream autonomous driving tasks. MagicDrive [19] presents a street-view synthesis framework with precise 3D controls (e.g., camera poses, road maps) using cross-view attention, improving 3D object detection and BEV segmentation. DreamDrive [43] further combines video diffusion with hybrid Gaussian scene representations to synthesize 4D scenes with 3D-consistent dynamic video rendering. Most recently, UniScene [33] proposes an occupancy-centric approach that unifies semantic, visual, and LiDAR data generation. However, existing DWMs overlook the scene understanding ability of the driving environment.

Large Language Models for Driving Large Language Models (LLMs) have demonstrated impressive generalization ability and extensive world knowledge across various tasks, including scene understanding [27, 58, 75, 78, 82], visual question answering (VQA), and both 2D and 3D visual grounding. DriveGPT4 [66] processes front-view video inputs to predict vehicle actions while providing natural language justifications via an LLM. DriveLM [50] leverages LLMs for graph-based VQA and end-to-end autonomous driving. NuInteract [77] further integrates large vision-language models (LVLMs) with a spatial processor using a set of learnable queries, trained on the large-scale NuInteract dataset containing over 1.5M multi-view image–language pairs, covering dense scene captioning and diverse interactive tasks. GaussianVLM [26] introduces a 3D Gaussian-based visual question answering (VQA) framework, where a SceneSplat-style variational autoencoder (VAE) is employed to directly encode 3D Gaussian scenes. Hermes [80] proposes a BEV-based world model that integrates scene representation with a Vision-Language Model (VLM) for joint scene understanding and generation, which is the most closely related work to ours. However, existing approaches typically rely on image, point cloud, or BEV representations. In contrast, our method leverages a 3D Gaussian scene representation [10] that achieves explicit spatial alignment between language features and 3D geometry, resulting in more accurate multimodal correspondence and improved representation of both texture and structural information in complex environments.

3. Method

In this paper, we propose GaussianDWM, a unified framework with 3D gaussian scene representation for driving scenarios understanding and generation. The pipeline of our method is illustrated in Fig. 2. The input to our method consists of images $\{I_i\}$, Gaussian ellipsoids $\{G_i\}$, and query text $\{t_i\}$. The framework is composed of three main modules:(i) World tokenizer (Sec. 3.1); (ii) Scene Understanding(Sec. 3.2); (iii) Multi-modal Generation (Sec. 3.3). We elaborate on the entire pipeline of our system in the following subsections.

3.1. World Tokenizer

Our world tokenizer encodes the world observations, i.e., the current multi-view images, into a compact continuous 3D Gaussian representation. We then apply a gaussian projector to align the selected 3D features to the text space before processing by the VLM.

3D Gaussian Tokenizer To preserve texture, 3D structure, and language alignment, we adopt 3D Gaussians as the scene representation for LLM input. We build upon LangSplat [49] to construct a 3DGS language field, where each Gaussian is augmented with a language embedding f_i. These embeddings are obtained from CLIP features, which inherit hierarchical semantics extracted via SAM [31]. We then follow the standard 3DGS rendering strategy, incorporating language information directly into the Gaussian primitives.

$$\boldsymbol{F}(v) = \sum_{i \in \mathcal{N}} f_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \tag{1}$$

where $\boldsymbol{F}(v)$ represents the language embedding rendered at pixel v, and $\alpha_i = o_i G_i^{2D}(v)$. Here o_i is the opacity of the i th Gaussian and $G_i^{2D}(\cdot)$ represents the function of the i-th Gaussian projected onto 2D. To further reduce memory consumption and improve efficiency, we introduce a scene-wise language autoencoder E, which maps the CLIP embeddings $\boldsymbol{F}(v) \in \mathbb{R}^D$ to $\boldsymbol{H}(v) = E(\boldsymbol{F}(v)) \in \mathbb{R}^d$, where $d \ll D$. We select $d = 3, D = 512$. Then we learn a decoder Ψ to reconstruction CLIP feature. Our autoencoder can significantly decrease memory requirements while retaining semantic fidelity.

3D Gaussian Projector We first align the extracted 3D Gaussian tokens to the text space. For each Gaussian primitive G_i, we represent its attributes as $G_i = (x_i, o_i, s_i, r_i, f_i)$ where $x_i \in \mathbb{R}^3$ denotes the 3D spatial position, o_i the opacity, s_i the scale, r_i the rotation, and f_i the associated CLIP feature. For the Gaussian tokenizer, we first apply learnable Fourier embeddings [44] to the 3D coordinates x_i:

$$\gamma(x_i) = \left[\sin(2^k \pi x_i), \cos(2^k \pi x_i)\right]_{k=0}^{L-1}, \tag{2}$$

where L is set to 10. For the opacity o_i, we apply a sigmoid activation $\hat{o}_i = \sigma(o_i)$ to constrain the value to $[0, 1]$. For the CLIP feature f_i, we use a pre-trained scene-wise decoder to project it to a 512 dimension $\tilde{f}_i = \Psi(f_i) \in \mathbb{R}^{N \times 512}$. N denotes the number of 3D Gaussian ellipsoids. Then, we employ a set of MLP projectors to map Gaussian attribute into a shared 4096-dimensional feature space, i.e., $h_i^x = \phi_x(\gamma(x_i))$, $h_i^o = \phi_o(\hat{o}_i)$, $h_i^s = \phi_s(s_i)$, $h_i^r = \phi_r(r_i)$, and $h_i^f = \phi_f(\tilde{f}_i)$, where $\gamma(\cdot)$ is the Fourier embedding and each $\phi_\cdot(\cdot)$ is a learnable MLP. Finally, we fuse the projected

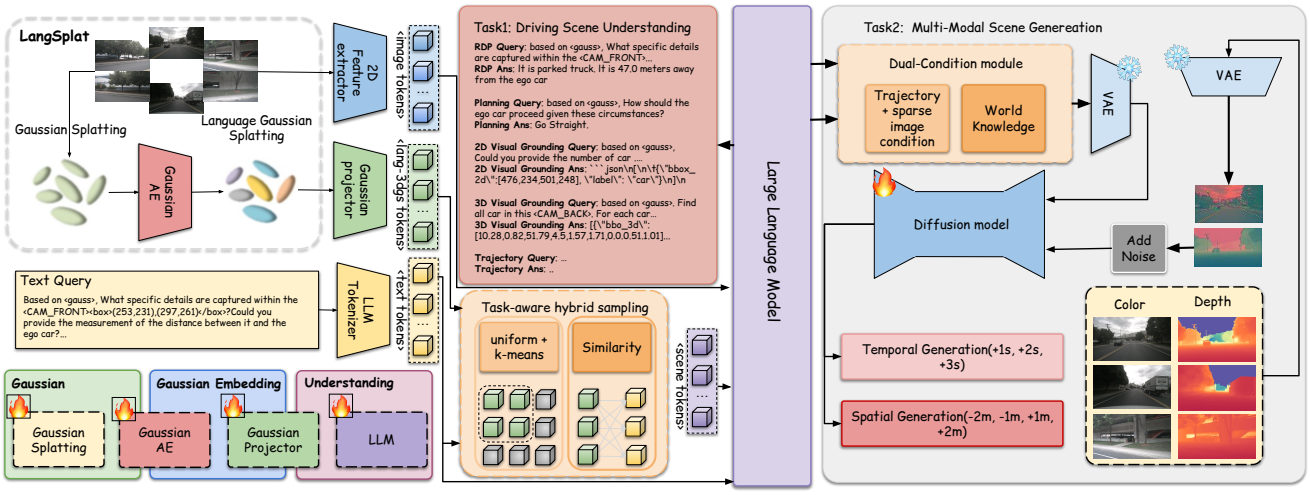


Figure 2. **System Overview.** We propose the first unified 3D Gaussian-based world model framework that simultaneously supports both scene understanding and scene generation. We first employ a scene encoder to align the language information with the 3D Gaussians, resulting in language-augmented 3D Gaussian representations. Then, a designed Gaussian projector aligns the 3D Gaussian tokens, 2D image tokens, and text tokens into a unified latent space. Subsequently, a task-aware hybrid sampling strategy is applied to select the most relevant 3D Gaussian tokens for the current query, which are then fed into the LLM. The LLM produces both textual answers and high-level language features that encapsulate world knowledge, which are later used to guide multi-modal scene generation.

features via learnable weights to obtain the Gaussian scene token $\mathcal{G}_i = \sum_{p\in\{x,o,s,r,f\}} \alpha_p \cdot h_i^p$, where each α_p is a trainable scalar normalized by a softmax. For text queries, we tokenize the input prompts into vocabulary indices and text tokens $\mathcal{T}$ for LLM processing.

3.2. Scene Understanding

This section introduces the world understanding module. The Large Language Model (LLM) interprets driving scenarios from the world tokenizer outputs $\mathcal{G}_i \in \mathbb{R}^{N\times C}$ according to user instructions. Then, the LLM parses the user instruction $\mathcal{T}_i$ and extracts world knowledge from the driving scene, generating both a textual answer t_i and a language feature representation C_L, which is later used as a condition signal for scene generation. This feature encodes high-level world knowledge as well as spatial information and is later used as a condition for scene generation. We implement the LLM using the widely adopted Qwen3 model [69]. The overall architecture is:

$$\{t_i, C_i^l\} = LLM(\mathcal{G}_i, \mathcal{T}_i) \tag{3}$$

Task-aware Language-guided Sampling However, directly converting all Gaussians into tokens would exceed the maximum token length limits of LLMs, and the high degree of redundancy in the Gaussian set would make it difficult for the LLM to reason about spatial interactions across different views. To address this, we propose a task-aware hybrid sampling strategy tailored for 3D Gaussian scene representations. For scene understanding tasks, we adopt a global sampling strategy that preserves holistic scene information by selecting a representative subset of Gaussians. We apply both uniform sampling and top-k sampling to select $N = 4096$ gaussian tokens from the hundreds of thousands in the scene, which are then fed into the LLM. In contrast, for 2D and 3D visual grounding tasks, we further introduce a language-guided sampling module, which re-tokenizes the dense scene representation into a more compact and sparse form conditioned on the text query. Specifically, we apply similarity calculation between the 3D Gaussian features and the text query to identify and retain only those Gaussians that are most relevant to the query. This design is highly effective for both 2D and 3D grounding tasks, as it enables the model to selectively inject the most relevant 3D spatial information into the language reasoning process. Compared with the previous Hermes framework, our 3D Gaussian–based LLM model can effectively respond to user queries about the driving environment, providing scene descriptions and answers to visual questions. In addition, it supports both 2D and 3D visual grounding tasks, achieving state-of-the-art performance in terms of averaged metrics.

Training Strategy Similar to many VLM training protocols, we adopt a two-stage training strategy consisting of an alignment stage and a fine-tuning stage. In the first stage, we freeze the entire VLM and train the aligner with full parameters for 5k warm-up steps to align the visual representations with the textual space. In the second stage, we further adapt the LLM using LoRA for 30k steps. Both stages share the same training objective. Following, we use a prefix language modeling, where the model is conditioned on an input prefix and trained to autoregressively generate the target continuation:

$$\mathcal{L}(\theta, \mathcal{B}) = - \sum_{\{t_{\text{prefix}}, t_{\text{gt}}\} \in B} \sum_{i=1}^{|t_{\text{gt}}|} \log p_\theta \left(t_{\text{gt}}^{(i)} \mid t_{\text{gt}}^{(<i)}, t_{\text{prefix}} \right), \tag{4}$$

where θ are the model parameters, $\mathcal{B}$ denotes a batch of samples of prefix input t_{prefix} (text tokens, image tokens and 3D Gaussian tokens), and ground truth response t_{gt}.$t_{\text{gt}}^{(t)}$ denotes the t-th token in the ground truth response sequence.

3.3. Multi-modal Scene Generation

In this section, we propose a dual condition multi-modal scene generation model. Our model consists a denoising UNet [1], and a frozen pre-trained VAE [64] to encode RGB images I_i, and depth maps D_i into a unified latent space. To satisfy the VAE input specifications, we convert depth maps into pseudo-RGB images via channel replication. The VAE encoding process can be written as: $z_I = \mathcal{E}(I_i), \quad z_D = \mathcal{E}(D_i)$ During decoding, the VAE decoder $\mathcal{D}$ reconstructs RGB, and depth from the latent variable z_i. For the depth map, we average the three output channels of the decoded result to obtain the final single-channel depth prediction: $\hat{I}_i = \mathcal{D}(z_I), \quad \hat{D}_i = \frac{1}{3}\sum_{c=1}^{3} \mathcal{D}(z_D)_c$ In the training phase, we random sample on the known camera trajectory, and get the latent code of corresponding color and depth images with the VAE encoder z_I, z_D. At each timestep t, we add noise to the sampled data. Then, we use the projection matrix to project surrounding point cloud at time t to time $t+n$ to serve as low-level image conditions $\{C_I, C_D\}$. Then we concatenate the noisy latent representations of each modality with the low-level image control signal and high-level language control signal C_L from LLM as the input to the denoising diffusion network. The model is using a v-prediction objective. The target $\mathbf{v}_t$ is defined as:$\mathbf{v}_t = \alpha_t \boldsymbol{\epsilon}_t - \sigma_t d_t$, where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, I)$ denotes the sampled gaussian noise, α_t and σ_t represent the time-dependent noise scheduling coefficients, and $\boldsymbol{d}_t$ corresponds to the noisy input modality requiring denoising. The training objective is defined as:

$$\mathcal{L} = \mathbb{E}_{d, \boldsymbol{\epsilon}, t, \boldsymbol{s}} \left\| \mathcal{F}_\theta \left(d_t, d_{\text{ref}}, C_I, C_D, C_L, \boldsymbol{s} \right) - \mathbf{v}_t \right\|_2^2, \tag{5}$$

The low-level conditions C_I and C_D represent the scene's texture and geometric information, guiding the generation process. Meanwhile, the high-level language feature C_L encapsulates comprehensive world knowledge extracted from the LLM. By conditioning the generation on multiple levels of information, our model achieves more accurate and consistent temporal and spatial synthesis.

For spatial generation, we project the surrounding point cloud using the spatial transformation of the query frame to obtain a sparse condition map. For temporal generation, we utilize the trajectory predicted by the front-end LLM to project the point cloud and construct a temporal sparse condition map, enabling temporally coherent scene generation.

Our generation model supports both spatial scene generation, i.e., novel view synthesis with spatial shifts of 1m or 2m, and temporal scene generation, i.e., future scene prediction at 1s and 2s into the future.

4. Experiments

4.1. Dataset and Evaluation Metric

We evaluate our method on two datasets for both scene understanding and scene generation. NuScenes [2] is a widely used autonomous driving dataset. We use six surrounding camera images as inputs. NuInteract [77] are recent benchmark for scene understanding. NuInteract provides ∼1.5M annotations and supports multiple tasks, including 2D perception and 3D visual grounding. For scene understanding, we adopt standard language and captioning metrics, including gCIDEr [54], BLEU-4 [46], ROUGE [39]. For scene generation, we use FID for images, and FVD for videos as evaluation metrics.

4.2. Implementation Details

Our training pipeline consists of three stages. In the first stage, we train the Gaussian tokenizer, projector, and the proposed sampling strategy independently. We then integrate these components with the LLM and perform joint fine-tuning. All models in this stage are trained using 16 NVIDIA A100 GPUs. In the second stage, we train the multi-modal generation module. We start by training a low-resolution (224×400) RGB video generation model, extend it to low-resolution RGB-D generation, and finally refine it into a high-resolution (424×800) RGB-D video generator using a mixed-frame-length strategy. The model is optimized using simulation-free rectified flow and a v-prediction loss [17]. In the final stage, we perform end-to-end joint optimization over all components to ensure consistency between scene understanding and scene generation.

4.3. Scene Understanding

In this section, we present the results of scene understanding, which are divided into four tasks: region description and perception, planning, 2D visual grounding, and 3D visual grounding. The results are shown in Tab. 1. The experimental settings strictly follow Drivemonkey [77]. We primarily compare two categories of methods: (1) LLM-based visual grounding approaches such as LLaVA [41] and InternVL [6], and (2) specialized 3D detection models such as BEVFormer [35], PETR [42], and CAPE [65]. As shown in Tab. 1, GaussianDWM significantly outperforms previous LVLMs in terms of average metrics across all tasks on the NuInteract test dataset. It achieves state-of-the-art performance in terms of averaged metrics on the RDP, 2D VG, 3D VG, and Planning tasks, outperforming previous state-of-the-art methods by a relative margin of 13.6%, ef-

Model	Years	LLM	2D RD & Pre ↑			2D VG ↑			3D VG ↑			Plan ↑	Avg. ↑
			BLEU	Rouge_L	CIDEr	mAP	F1	MIoU	Pr	mAP	F1	Acc	
LLaVA1.5	2024	Vicuna-7B	64.23	76.69	74.82	0.10	0.16	14.31	6.51	5.33	3.12	36.20	28.16
MiniCPM-V 2	2024	MiniCPM-2B	47.43	63.16	69.88	0.11	0.13	13.34	0.97	1.55	0.86	36.69	23.41
MiniCPM-V 2.6	2024	QWen2-7B	47.92	69.11	70.20	0.36	0.49	18.74	1.97	1.61	0.93	36.42	24.78
InternVL1.5-2B	2024	InternLM2-7B	67.14	81.10	79.83	14.74	17.64	55.43	28.05	21.73	12.92	53.96	43.25
InternVL1.5-4B	2024	Phi3-4B	66.63	80.64	79.24	14.27	17.60	53.52	25.14	19.46	11.63	40.25	40.84
QWen2VL	2024	Qwen2-2B	67.92	80.24	78.51	17.11	20.87	57.24	12.82	10.20	6.12	45.59	39.66
Qwen2VL	2024	QWen2-7B	66.65	78.57	77.97	16.06	20.04	55.51	20.64	16.26	9.82	49.33	41.09
InternVL2-1B	2024	Qwen2-0.5B	66.89	81.00	79.59	16.70	20.21	55.94	23.36	18.35	10.94	44.08	41.71
InternVL2-2B	2024	InternLM2-2B	66.77	80.87	79.62	16.12	19.49	55.29	27.83	21.09	12.58	44.61	42.43
InternVL2-4B	2024	Phi3-4B	66.88	80.76	79.29	19.14	23.47	59.07	25.28	20.12	11.97	40.43	42.64
InternVL2-8B	2024	InternLM2.5-7B	67.32	**81.39**	**80.01**	20.61	25.24	61.90	31.47	24.67	14.70	46.93	45.42
DriveMonkey	2025	InternLM2.5-7B	67.50	81.15	79.79	19.47	24.02	59.36	51.90	34.53	20.86	**82.64**	52.12
Bevformer	2022	-							44.50	23.69	1.576		
PETR	2022	-			Unsupported				**55.80**	31.34	20.58	Unsupported	
CAPE	2023	-							55.02	32.94	21.33		
GaussianDWM	2025	Qwen3-8B	**68.78**	81.06	78.72	**34.95**	**40.49**	**71.85**	50.66	**52.78**	**32.05**	80.95	**59.23**

Table 1. The comparison between our **GaussianDWM** and other state-of-the-art models on the NuInteract dataset [77]. The scene understanding task includes four subtasks: region description and perception, 2D visual grounding, 3D visual grounding, and planning. Our method achieves state-of-the-art average performance across all four tasks, which fully demonstrates the effectiveness of introducing a 3D Gaussian scene representation for enhancing the LLM's capability to understand 3D spatial information.

Finetune	Gaussian	Sampling	2D RD & Pre ↑			2D VG↑			3D VG↑			Plan↑	Avg.↑
			BLEU	Rouge_L	CIDEr	mAP	F1	MIoU	Pr	mAP	F1	Acc	
zeroshot	w/o		2.91	12.68	0.59	0.00	0.00	12.24	48.75	47.59	29.12	0.00	15.39
finetuned	w/o		65.09	78.35	76.56	30.01	35.45	69.01	50.36	51.88	31.43	45.09	53.32
finetuned	w	Random	66.19	79.00	76.97	33.94	39.37	71.40	50.94	52.85	32.03	49.43	55.21
finetuned	w	Top-k + Uniform	**68.78**	**81.06**	**78.82**	33.89	39.31	71.37	**51.16**	**52.87**	**32.05**	**80.95**	58.93
finetuned	w	Top-k + Uniform + similarity	**68.78**	**81.06**	**78.82**	**34.95**	**40.49**	**71.85**	50.66	52.78	**32.05**	**80.95**	**59.23**

Table 2. We conduct ablation studies to validate the effectiveness of each proposed component, including the 3D Gaussian scene representation, the top-k and uniform sampling strategies, and the similarity–based sampling module. Note that the similarity sampling strategy is applied only to grounding tasks requiring focused attention (e.g., 2DVG and 3DVG).

fectively demonstrating the importance of introducing a 3D Gaussian scene representation for enabling LLMs to better comprehend 3D spatial information. Compared with previous point cloud– and BEV-based scene representations, our 3D Gaussian formulation allows for explicit alignment between 3D structural, texture, and language features, leading to more efficient and semantically consistent information injection into LLMs. Moreover, our task-aware hybrid sampling strategy efficiently selects Gaussian tokens most relevant to the query text while filtering out redundant 3D Gaussians. Compared with the current state-of-the-art VQA method Drivemonkey, our model shows clear advantages in both 2D and 3D visual grounding tasks, while also achieving comparable performance to specialized 3D detectors designed specifically for 3D VG.

4.4. Scene Generation

In this section, we present the evaluation of multi-modal scene generation, which includes both spatial scene generation and temporal scene generation. All experiments are conducted on the nuScenes dataset [2]. Following previous works [19, 25], we interpolate the 2Hz keyframe annotations to a higher frame rate of 12Hz, and the experimental settings strictly follow [25].

We compare our method with representative street-view synthesis approaches, including PVG [4], EmerNeRF [71], StreetGaussian [68], OmniRe [7], FreeVS [55], DiST-4D [24]. As shown in Tab. 3, our method outperforms all existing methods and achieves state-of-the-art performance. We use FID and FVD as evaluation metrics since ground-truth data is not available after spatial shifts. Our method demonstrates superior consistency and photorealism under extreme viewpoint shifts ($\pm$4m), outperforming direct reconstruction-based approaches. This indicates that our framework effectively combines the advantages of 3D Gaussian scene representation and diffusion-based generative modeling. Under the guidance of our dual-condition mechanism—leveraging both high-level world knowledge and low-level geometric cues—our method achieves state-of-the-art spatial and texture fidelity under large viewpoint variations.

We also visualize the qualitative results of spatial scene generation in Fig. 3. We further compare our method with several existing novel view synthesis approaches, and the qualitative results are presented in Fig. 4. As shown, our RGB and depth generations exhibit photorealistic qual-



Figure 3. **Qualitative results for scene understanding and scene generation**. From top to bottom, we display the multi-view input of the current scene and the 3D Gaussian ellipsoids, the scene understanding results, and the spatial and temporal scene generation results.

ity, demonstrating the complementary strengths of world knowledge, diffusion-based generation, and 3D Gaussian scene representation.

The proposed dual-condition design enables high-level world knowledge to effectively guide the model's temporal reasoning for future scene prediction, while the low-level sparse condition constrains 2D texture and style consistency. This demonstrates the importance of incorporating explicit **world knowledge** into world models to improve both semantic and spatial coherence during temporal generation. Overall, GaussianDWM is capable of simultaneously understanding and generating complex driving scenarios, highlighting a promising research direction toward unified scene understanding and generation in world modeling.

4.5. Ablation Study

In this section, we conduct comprehensive ablation studies to validate the effectiveness of each component in our proposed framework. We systematically analyze the im-

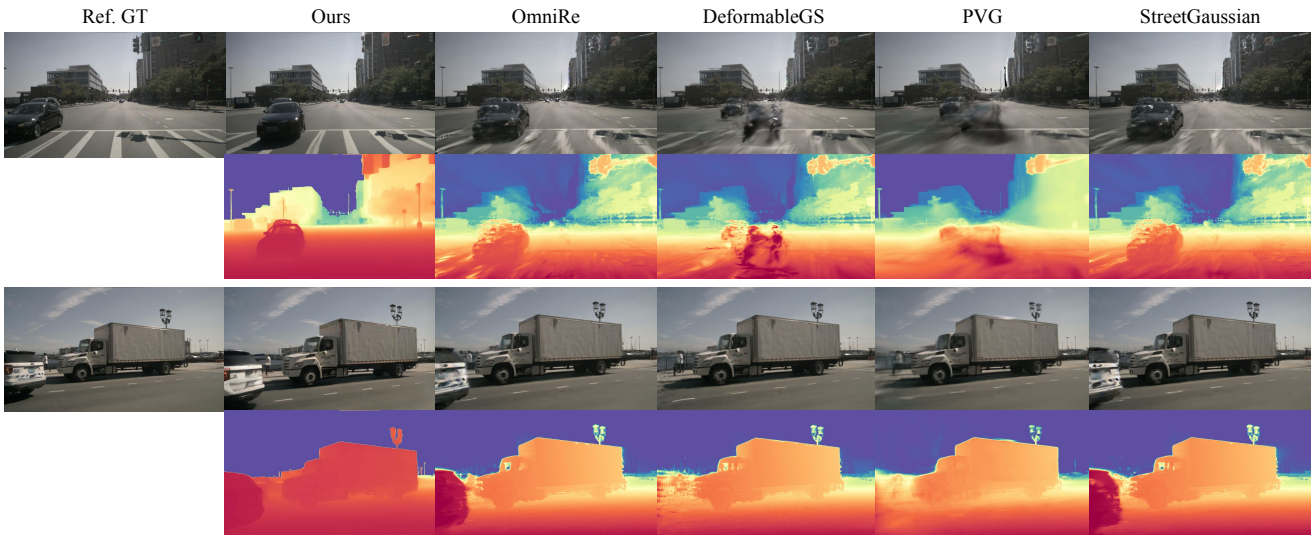


Figure 4. **Qualitative comparison of RGB-D NVS with 2m shift**. Compared with state-of-the-art reconstruction-based methods for spatial NVS [4, 7, 68, 73], our method reduce artifacts of dynamic objects and preserves temporal-spatial consistency across large viewpoint shifts.

Method	Shift ± 1		Shift ± 2		Shift ± 4	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
PVG	48.15	246.74	60.44	356.23	84.50	501.16
EmerNeRF	37.57	171.47	52.03	294.55	76.11	497.85
StreetGaussian	32.12	153.45	43.24	256.91	67.44	429.98
OmniRe	31.48	152.01	43.31	254.52	67.36	428.20
FreeVS	51.26	431.99	62.04	497.37	77.14	556.14
DiST-S	10.12	45.14	12.97	68.80	**17.57**	**105.29**
Ours	**8.36**	**44.50**	**11.27**	**68.17**	18.81	116.40

Table 3. **Quantitive comparison** between our method and other novel view synthesis methods on nuScenes dataset.

pact of the key modules, including the 3D Gaussian scene representation, the hybrid sampling strategies, and the dual-condition generation design, to demonstrate their respective contributions to the overall performance.

3D Gaussian Representation Compared with other scene representations, the 3D Gaussian scene representation provides a better ability to encode environmental features, including both texture and geometric information, and can be directly aligned to the 3D space through language features. The results are shown in Tab. 2. Through ablation studies, we fully verify that the introduction of 3D Gaussian representations enhances the LLM's 3D scene understanding capability, enabling it to better perceive the geometric and semantic information of the scene.

Hybrid Sampling Based on the observed redundancy in the 3D Gaussian scene representation, we design a hybrid sampling strategy to extract the most informative components of the environment. The results are shown in Tab. 2. By comparing random sampling, uniform+top-k sampling, and similarity-based sampling, we observe that when similarity-based sampling is applied between the text query and the 3D Gaussian tokens, the model can more effectively select task-relevant Gaussian tokens, leading to improved scene understanding performance.

Low-Level	High-level	FID±1m	FID±2m
✗	✓	-	-
✓	✗	10.12	45.14
✓	✓	**8.36**	**44.5**

Table 4. **Ablation Study of dual-condition generation mechanism**. "–" denotes failure under the setting.

High-level World Knowledge Compared with other world model and scene generation frameworks, we design a dual-condition guidance framework that controls scene generation at different levels. The ablation results for this part are shown in Tab. 4. It can be observed that our method performs better in long-sequence prediction and under large viewpoint variance, demonstrating the effectiveness of incorporating world knowledge.

5. Conclusion

This paper introduces a novel unified driving world model framework that supports both scene understanding and generation within a single architecture. We bridge the gap between these two tasks by proposing a new 3D Gaussian scene representation combined with a task-aware hybrid Gaussian sampling strategy, enabling effective world-query injection into the LLM. Extensive experiments validate the effectiveness of the proposed framework, demonstrating significant improvements in both scene understanding and generation. We believe this work an important step toward unified driving world models with 3D Gaussian scene representations.

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GaussianDWM: 3D Gaussian Driving World Model for Scene Understanding and Multi-Modal Generation (Supplementary Material)

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A. Overview

In this supplementary material, we first provide additional details on constructing the 3DGS dataset from the large-scale nuScenes dataset, as well as the caption dataset used for trajectory prediction. We then describe the implementation and training details in Sec. C and Sec. D. Finally, in Sec. E, we present extensive experimental results across various datasets.

B. Dataset Details

To represent scenes from the NuScenes dataset [2] using 3D Gaussians, we construct the first large-scale 3D Gaussian dataset for outdoor urban environments, comprising 800 scenes derived from the original nuScenes sequences. Our goal is to build a dataset that enables generalizable 3D Gaussian–based scene understanding with LLMs, providing high-quality geometry and appearance representations suitable for downstream reasoning tasks.

Dataset Processing Many scenes contain frames with very low overlap or severe motion blur, making it difficult to obtain accurate 3DGS reconstructions. Therefore, we remove images or entire scenes whose reconstruction quality does not meet our standard. Since NuScenes dataset [2] provides high-quality LiDAR point clouds, we initialize each scene using aggregated LiDAR points, avoiding the large-scale inaccuracies that would arise from using COLMAP as in conventional 3DGS pipelines. For scenes with available depth supervision, we additionally apply a depth loss to improve geometric fidelity. After optimization, we further filter reconstructed scenes based on the PSNR metric before using them as inputs for our pre-training.

Language Label Collection and Training Our language label collection aims to establish 3D language paired data by associating each 3D Gaussian primitive with a rich language feature f_i. We build upon LangSplat [6] to construct a 3DGS language field. These embeddings are obtained from CLIP features, which inherit hierarchical semantics extracted via SAM [4]. We then follow the standard 3DGS rendering strategy, incorporating language information directly into the Gaussian primitives. As a collection of anisotropic 3D Gaussians, with each Gaussian $G(x)$ characterized by a mean $\mu \in \mathbb{R}^3$ and a covariance matrix Σ:

$$G(x) = \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right). \tag{1}$$

To optimize the parameters of 3D Gaussians, they are rendered into 2D image planes and a tile-based rasterizer is used to improve the rendering efficiency:

$$C(v) = \sum_{i\in\mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1}(1-\alpha_j), \tag{2}$$

We also adopt the tilebased rasterizer to retain the rendering efficiency:

$$\boldsymbol{F}(v) = \sum_{i\in\mathcal{N}} f_i \alpha_i \prod_{j=1}^{i-1}(1-\alpha_j) \tag{3}$$

where $\boldsymbol{F}(v)$ represents the language embedding rendered at pixel v, and $\alpha_i = o_i G_i^{2D}(v)$. Here o_i is the opacity of the i th Gaussian and $G_i^{2D}(\cdot)$ represents the function of the i-th Gaussian projected onto 2D.

For scene representation, each environment typically requires several million Gaussians to model its geometry and appearance. Since CLIP embeddings are high-dimensional features, directly learning the latent feature f_i for every Gaussian in the CLIP space would dramatically increase both memory consumption and computational cost—particularly because each f_i has a dimensionality of 512.

To further reduce memory consumption and improve efficiency, we introduce a scene-wise language autoencoder E, which maps the CLIP embeddings $\boldsymbol{F}(v) \in \mathbb{R}^D$ to $\boldsymbol{H}(v) = E(\boldsymbol{F}(v)) \in \mathbb{R}^d$, where $d \ll D$. We select

$d = 3, D = 512$. The autoencoder is trained with a reconstruction objective on the CLIP embeddings. We also learn a decoder Ψ to reconstruct CLIP feature. Our autoencoder can significantly decrease memory requirements while retaining semantic fidelity.

$$\mathcal{L}_{ae} = \sum_{t=1}^{T} d_{ae} \left(\Psi \left(E \left(\boldsymbol{F}(\boldsymbol{v}) \right) \right), \boldsymbol{F}(v) \right), \tag{4}$$

After training the autoencoder, we transform all CLIP embeddings into scene-specific latent features $\boldsymbol{H}(v)$. We let our 3D language Gaussians learn language embeddings in the scene-specific latent space instead of the CLIP latent space. Compared to directly modeling the 512-dimensional CLIP embeddings, we significantly reduced the memory cost by incorporating scene priors. We optimized the language embeddings with the objective:

$$\mathcal{L}_{\text{lang}} = \sum_{l \in \{s,p,w\}} \sum_{t=1}^{T} d_{\text{lang}} \left(\boldsymbol{F}_t^l(v), \boldsymbol{H}_t^l(v) \right) \tag{5}$$

where $d_{\text{lang}}()$ denotes the distance function used for our 3D Language Gaussians. To inject the 3D Gaussian tokens into the query text input, we reconstruct a dedicated QA dataset. Moreover, to align with the training paradigm of QwenVL, all 3D Gaussian token annotations are placed at the beginning of each instruction. The exact formatting protocol is illustrated as follow:


```
{
  "token": "fd8420396768425eabec9bdddf7e64b6",
  "scene_token": "e7ef871f77f44331aefdebc24ec034b7",
  "scene_idx": 2,
  "frame_idx": 0,
  "category": "2D_perception_infos",
  "task": "rdp",
  "conversations": [
    {
      "from": "human",
      "value": "based on <gauss>, Can you quantify the distance separating the <CAM_FRONT> <box>(14,219),(37,248)</box> from the ego car?"
    },
    {
      "from": "gpt",
      "value": "38.7 meters is the distance from the ego car to it"
    }
  ],
  "image": [
    "nuscenes/samples/CAM_FRONT/n015-2018-08-02-17-16-37+0800__CAM_FRONT__1533201470412460.jpg"
  ],
  "views": ["CAM_FRONT"],
  "gauss": [
    "gauss/output-full-6v/2_CAM_FRONT/langsplat_3/per_frame/00000.pth"
  ]
}
```


Trajectory Dataset Generation Our generative model supports both temporal and spatial scene generation. For spatial generation, text queries typically take the form "generate the view after shifting 2 m to the left." In this case, we first use the LLM to infer the target camera pose. Using the predicted pose, we project the point cloud to obtain a sparse condition, which serves as the low-level guidance for the diffusion model.

For temporal generation, queries are typically phrased as "predict the scene 1 second later." The LLM performs trajectory prediction and directly outputs the future camera pose. Based on this pose, we again apply point-cloud projection to generate the sparse condition used for temporal guidance.

To enable temporal scene generation, the LLM must possess the capability to predict future trajectories. Based on this requirement, we construct a trajectory-oriented QA dataset to supervise and enhance the model's trajectory reasoning ability. From each video clip, we extract a 10-frame sequence. The first 4 frames are provided to the model as the prompt input, while the remaining 6 frames are used as the prediction targets. All trajectories are transformed into the ego-coordinate system of the 5th frame. The exact formatting protocol is illustrated as follow:


```

    "conversations": [
        {
            "from": "human",
            "value": "There is last 4frames trajectory, [PT, [-7.45, 3.05, 0.16, 0.0, 0.0, -0.36, 0.93], [-5.75, 1.97, 0.11, 0.0, 0.0, -0.28, 0.96], [-4.04, 0.92, 0.09, 0.0, 0.0, -0.19, 0.98], [-2.15, 0.26, 0.04, 0.0, 0.0, -0.1, 1.0]]. Summarize the motion of the ego vehicle in this 6-frame clip"
        },
        {
            "from": "gpt",
            "value": "[PT, [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0], [2.47, 0.65, -0.06, 0.0, 0.0, 0.09, 1.0], [4.24, 2.32, -0.11, 0.0, 0.0, 0.12, 0.99], [5.5, 4.08, -0.15, 0.0, 0.0, 0.13, 0.99], [7.03, 6.45, -0.15, 0.0, 0.0, 0.13, 0.99], [8.66, 8.91, -0.17, -0.01, 0.0, 0.13, 0.99]]"
        }
    ]
```


C. Implementation Details

Our training pipeline consists of three stages. In the first stage, we train the Gaussian tokenizer, projector, and the proposed sampling strategy independently. We then inte-

grate these components with the LLM and perform joint fine-tuning. For the LLM, we adopt a LoRA-based fine-tuning strategy. All models in this stage are trained using 16 NVIDIA A100 GPUs. In the second stage, we train the multi-modal generation module. We start by training a low-resolution (224×400) RGB video generation model, extend it to low-resolution RGB-D generation, and finally refine it into a high-resolution (424×800) RGB-D video generator using a mixed-frame-length strategy. The model is optimized using simulation-free rectified flow and a v-prediction loss [3]. In the final stage, we perform end-to-end joint optimization over all components to ensure consistency between scene understanding and scene generation.

3D Gaussian Scene Loading Strategy. For each processed 3D Gaussian scene, the loading procedure depends on the type of Gaussian features associated with the given QA sample. Since each QA item focuses on a specific keyframe while the number of available camera views varies across scenes, we adopt the following unified strategy:

- **Surround-view Gaussian features.** When the QA sample requires panoramic information, we load the Gaussian features from all six surrounding views. The Gaussian tokens from all views are concatenated along the token dimension to form a unified set of scene tokens.
- **Single-view Gaussian features.** When the QA sample is associated with a specific camera view, we load only the Gaussian features of that view, without any concatenation.

This design ensures flexibility for different QA formats while maintaining consistent integration of 3D Gaussian scene information into the language model input.

Training Schedule and Parameter Efficiency. During training, the number of epochs is adapted to each task, as the difficulty and convergence behavior differ across region description and perception (RDP), 2D visual grounding (2DVG), 3D visual grounding (3DVG), and planning. Specifically, each task is trained for a task-dependent number of epochs to ensure stable convergence.

In the first-stage training, the number of trainable parameters accounts for only **0.45%** of the total model parameters, focusing mainly on the Gaussian tokenizer, projector, and sampling modules. During the LoRA-based fine-tuning stage, the trainable parameters account for merely **0.79%** of the total model parameters. This demonstrates that our framework achieves strong performance while maintaining high parameter efficiency.

World Knowledge Injection. In our framework, world knowledge is injected into the generative model through a text-conditioned encoder pathway. Specifically, we employ a pretrained language model to convert a natural-language caption into a sequence of semantic tokens, which serve as high-level conditioning signals for the diffusion UNet. Given an input caption, we first tokenize it and encode the sequence using a CLIP-based text encoder. This produces a set of contextualized text embeddings $E_{\text{text}} \in \mathbb{R}^{L \times D}$ where each token represents a semantic unit such as an object category, attribute, or action. Unlike pooled language vectors, which collapse all semantics into a single global descriptor, token-level embeddings preserve fine-grained, word-level structure and allow the diffusion model to selectively attend to relevant parts of the caption. To integrate text knowledge into the generative process, we concatenate the text embeddings with the image-embedding tokens extracted from a reference frame. We further append a learnable type embedding to each token, enabling the model to distinguish between visual and linguistic information during cross-attention. The resulting combined sequence $E_{\text{enc}} = [E_{\text{img}}; E_{\text{text}}] + E_{\text{type}}$ is passed as the encoder-hidden-states to the diffusion UNet, which uses cross-attention at multiple layers to fuse world knowledge with spatial features. Through this mechanism, the UNet learns to associate local visual structures with global textual semantics, enabling the model to leverage descriptions such as object identity, scene context, and commonsense priors to guide novel-view synthesis. This fusion pathway allows the generative model to incorporate externally grounded world knowledge—captured implicitly by large-scale text–image pre-training of the language encoder—without modifying the underlying UNet architecture. As a result, the model benefits from both low-level geometric cues (RGB/depth latents) and high-level semantic reasoning (text-conditioned cross-attention), improving structural consistency, scene understanding, and semantic controllability during view synthesis.

D. Ablation Study

We additionally provide further ablation studies in Tab. 1, including experiments on the effectiveness of the cross-attention–based sampling module. We also analyse on how world knowledge influences the performance of the generative model in Tab. 3.

E. Extensive Experiments

OmniDrive dataset [7] Scene Understanding Results. We further evaluate our model on the OmniDrive dataset, comparing against strong baselines including GPT-4o [1], LLaVA [5], OmniDrive [7], and Hermes [8]. Our GaussianDWM consistently achieves state-of-the-art performance across all evaluation metrics. These results again confirm the effectiveness of our 3D Gaussian scene representation and task-aware sampling strategy in enhancing LLM-based scene understanding under complex driving scenarios.

We also provide additional qualitative results, including more visualizations of the QA outputs and scene generation

Sampling	Gaussian Tokens	2D RD & Pre ↑			2D VG↑			3D VG↑			Plan↑	Avg.↑
		BLEU	Rouge_L	CIDEr	mAP	F1	MIoU	Pr	mAP	F1	Acc	
Top-k + Uniform+Cross	128	68.27	80.95	78.73	34.79	40.3	71.79	**51.41**	54.01	32.81	68.43	58.15
Top-k + Uniform+Cross	512	**68.37**	**81.00**	**79.22**	**34.95**	**40.39**	**71.85**	51.30	**54.07**	**32.87**	**68.52**	**58.26**
Top-k + Uniform	4096	66.17	79.07	77.06	33.89	39.31	71.37	51.16	52.87	32.05	66.27	56.92
Top-k + Uniform+Cross	4096	68.20	80.90	78.93	34.50	40.00	71.57	51.20	53.59	32.53	65.60	57.70

Table 1. **Ablation study on Sampling Strategies and Gaussian Token Counts.** We investigate the effectiveness of the proposed Cross-Attention sampling compared to the Uniform baseline. The model with 512 tokens using our strategy achieves the optimal performance, surpassing the dense baseline using 4096 tokens. This indicates that our learned sampling effectively captures critical information with significantly fewer tokens, validating its efficiency.

Model	Reference	# LLM Params	Understanding↑		
			METEOR	ROUGE_L	CIDEr
Only Understanding					
GPT-4o [1]	-	-	-	0.223	0.244
LLaVA-OV [5]	arXiv 24	7B	-	0.221	0.284
OmniDrive [7]	CVPR 25	7B	0.380	0.326	0.686
OmniDrive-2D [7]	CVPR 25	7B	0.383	0.325	0.671
OmniDrive-BEV [7]	CVPR 25	7B	0.356	0.278	0.595
Unified Understanding and Generation					
Separated unification	-	1.8B	0.384	0.327	**0.745**
HERMES [8]	ICCV 25	1.8B	0.384	0.327	0.741
GaussianDWM	-	8B	**0.395**	**0.341**	0.712

Table 2. **OmniDrive dataset [7] Scene Understanding Results.** We further evaluate our model on the OmniDrive dataset, comparing against strong baselines including GPT-4o [1], LLaVA [5], OmniDrive [7], and Hermes [8]. Our GaussianDWM consistently achieves state-of-the-art performance across all evaluation metrics. These results again confirm the effectiveness of our 3D Gaussian scene representation and task-aware sampling strategy in enhancing LLM-based scene understanding under complex driving scenarios.

results.

World Knowledge	Gaussian	shift ± 1m		shift ± 2m		shift ± 4m	
		FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
✗	✗	11.44	42.22	14.75	71.98	21.79	**137.31**
✓	✗	11.30	41.63	13.96	**70.45**	19.02	151.19
✓	✓	**11.28**	**41.62**	**13.95**	70.70	**18.91**	150.05

Table 3. **Ablation on World Knowledge and Gaussian**. We further evaluate the impact of world knowledge by comparing generation model with and without world knowledge, and by examining how Gaussian influences both world knowledge representation and final generation. The improvement brought by world knowledge becomes increasingly significant under larger view shifts. All evaluations are conducted on a subset of 30 scenes from the nuScenes dataset.

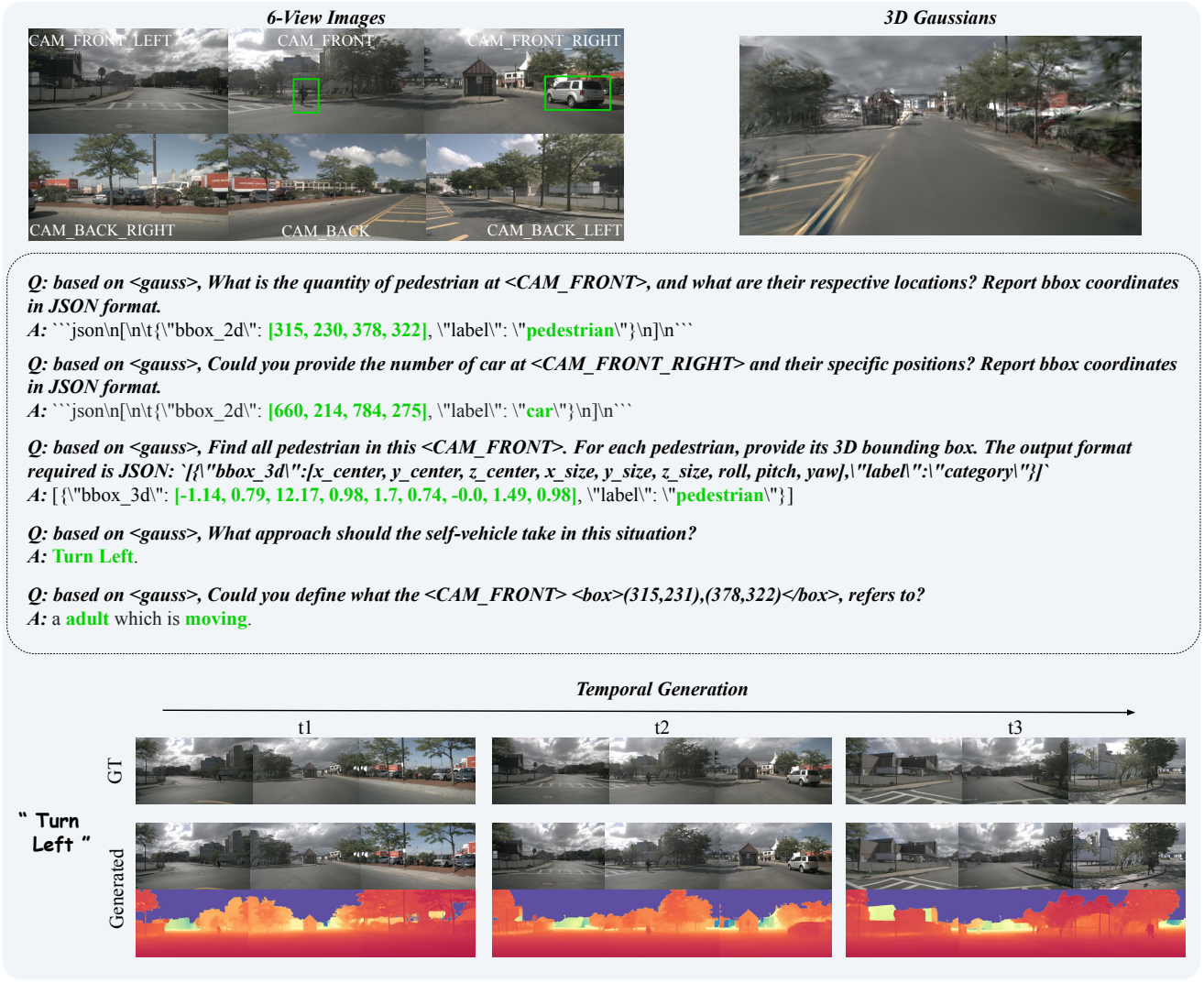


Figure 1. **Qualitative results for scene understanding and scene generation**. From top to bottom, we present the multi-view input of the current scene and the 3D Gaussian ellipsoids, the scene understanding results, and the spatial and temporal scene generation results.



Figure 2. **Qualitative results in rainy scene for understanding and generation**.

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